

		2
3	A	Kinetic energy of arrow $E = \frac{1}{2}mv^2 \text{ --- (1)}$ Momentum of arrow $p = mv \text{ --- (2)}$ $p^2 = 2mE$ $p = \sqrt{2mE}$ By Newton's second law, $F = \frac{p}{t}$ $Ft = p$ $= \sqrt{2mE}$ Same arrow, same m, different E,

Qn	Answer	Solution
		$Ft \propto \sqrt{E}$
4	C	Elastic collision $\Rightarrow$ by relative speed of approach = relative speed of separation